Shaft Conveyance

26/08/2019

Conveyance & Accessories I

*Conveyance used in mines are classified according to their use.

*Conveyance, handling personnel & material is called cage.

*Cage is used to lower & raise personnel, material & equipment.

*Conveyance handling broken ore or coal, waste rock are termed as skip.

*Combination of cage-skip are used at some places.

*Counter weight is also be considered as conveyance.

Conveyance & Accessories II

*Counter weight is used to balance with either skip or cage.

*Accessories consists of loading pockets, spill pockets & arrestor gear.

*Loading pocket improves efficiency of hoisting system by ensuring that optimum amount of material is fed to skip during each cycle.

*Arrestor gear is provided in head frame & shaft bottom for friction hoisting system to protect personnel, hoist & conveyance in event of over travel occurs at end of hoisting cycle.

Skip Hoisting I

*Skip hoisting consists of filling, hoisting, dumping & returning to be filled .

*Ore & waste may be crushed or uncrushed.

*Filling can be by volumetric measurement or weight.

*Filling operation can be manual or automatically controlled.

*During filling, skip is subjected to high dynamic loads & depending on depth , certain amount or rope stretching occurs.

Skip Hoisting II

*Chairing of skip is not done, due to higher impact loading with slack rope, danger of leaving chair in shaft which could cause collision.

*Skip is guided in shaft to avoid collision with shaft wall or other conveyance in shaft.

*Misalignment or loose guide results lower hoisting speed, thus reducing production of mine.

*Skip can dump ore at any point in shaft, most common location for dumping is head frame,

*Dumping is accompanied by means of scrolled located at each side of skip compartment at dump.

Skip Hoisting III

*A roller on side of skip body (for swing-out body skip) or on door (for fixed body skip) engages scroll & causes skip to dump ore.

*In some type of skip (Sala) an air cylinder actuates dump door.

Over Turning Skip I

*Over turning skip consists of bail frame, a shaft to support skip, skip body.

*Skip body rest on shaft.

*Skip is dumped through action of bull wheel, mounted on each side (at top) of skip body, emerging a scroll & causing skip body to rotate around shaft as skip is being hoisted through dump.

*Pay load is being discharged as skip moves along path.

Over Turning Skip II

Advantages

*There is less shaft spillage when handling fine or wet material.

*Handles larger pieces for a fixed cross sectional area.

Disadvantages

*Capacity is limited due to height to width ratio.

*Pay load ratio is lower particularly with sticky material,

*Sticky material tends to build up at bottom of skip.

*Generally impose higher stress on head frame bin.

*Requiring more head frame height to dump.

Swing Out Skip I

*Swing out body skip consists of bail frame, a shaft & a skip body.

*Skip body is suspended from shaft .

*Skip is dumped by means of roller mounted on each side (at bottom) of body.

*This engages a scroll & cause body to pivot out away from bail frame in to dump.

*At same time, a discharge door, which form bottom of skip opens & material is discharged.

*Safety latch is used to prevent opening of bottom door accidently, particularly when travelling empty.

Swing Out Skip II

*Cross section of skip decides size of material to be handled.

*Door opening size should be at least three times of largest piece to be handled.

*This prevents possible bridging.

*Length of skip, available space, hoist rope & centre to centre distance are to be considered before designing skip.

Swing Out Skip III

<u>Advantages</u>

*This have favorable pay load ratio then over turning skip.

*Tripping path is less complex than for over turning skip.

<u>Disadvantages</u>

*Muck build up at hinge point pin can prevent door from closing.

*This shall place large force on head frame scroll or skip body during closing cycle.

Swing Out Skip IV

*With wet material, these could be spillage from door leap.

*Bottom door mechanism is more complex than fixed body type.

Swing Out Body Skip I

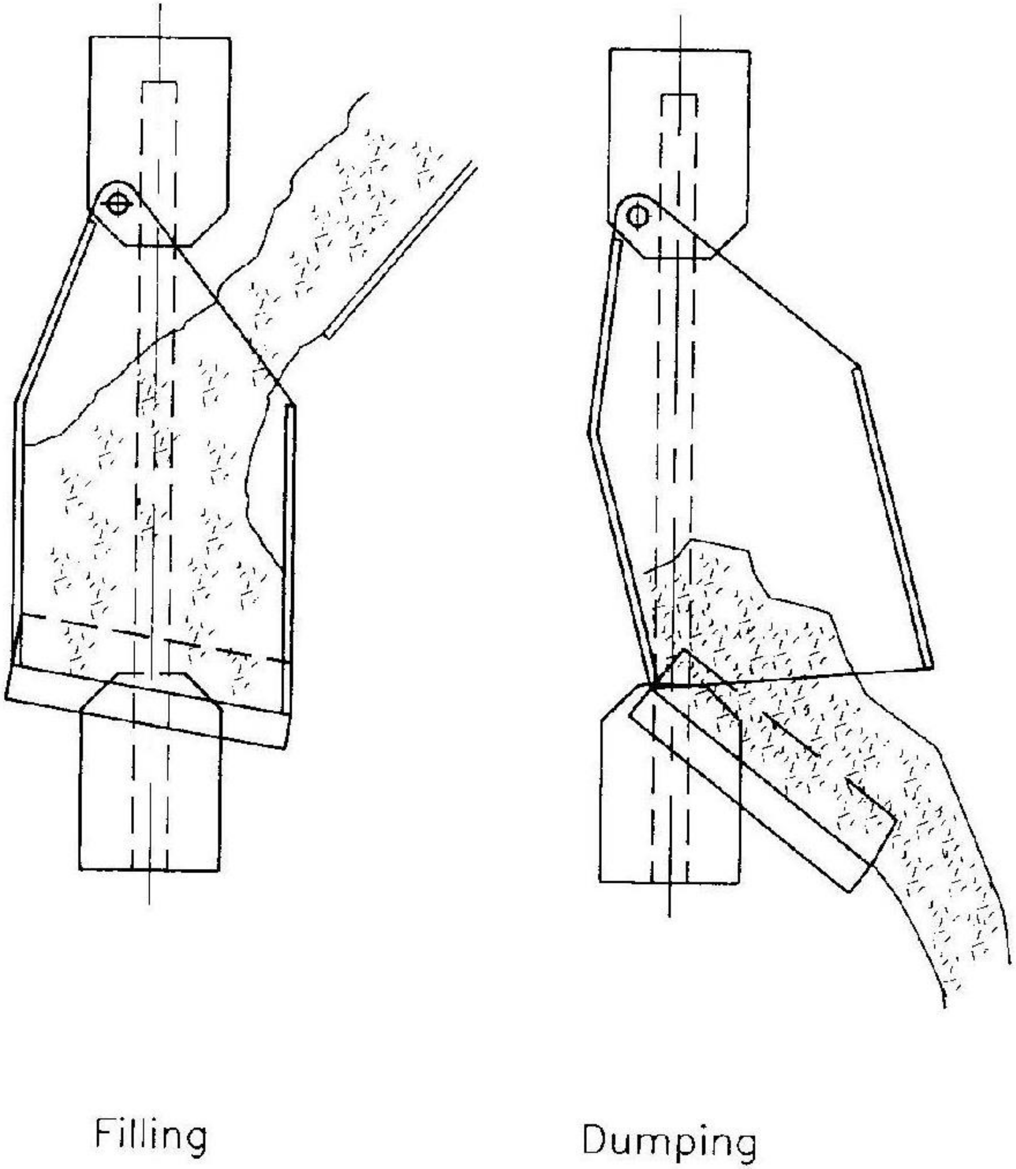

Fig. 17.5.8. Bottom dump skip, swing-out body (Penning, et al., 1977).

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Fixed Body Skip I

*body of skip remains fixed & dies not move out of confines of skip compartment during dumping.

*A door at bottom of body swing open to from a chute & allows material to discharge in to dump bin.

*This type of skip may or may not have bail.

*In fixed body (Rolla-Chute) skip, door is held closed by means of door latch.

*When skip enters head frame a latch track releases latch, allowing door to open & material to be discharge.

Fixed Body Skip II

*Arc-Door chute, a roller is mounted on each side of door.

*When skip enters skip dumps, roller engages a scroll, causing door to swing open.

*Sala skip, door is operated by means of compressed air cylinder mounted on skip body.

*Compressed air is to operated cylinder.

*Compressed air is supplied through a yoke, which is permanently mounted at head frame.

*Yoke is picked up by skip as it enters dumping area.

Fixed Body Skip III

*Cross sectional area of fixed body skip is determined by size of material to be hoisted.

*Fixed body skip can be built with out a bail frame.

*Cross section of skip can be increased by 10 to 15%, in comparison with skip with bail frame.

Advantages

*Skip can load & dump through guide.

*Skip has no bail frame, thus skip can be of larger cross section.

*This helps to reduce length of skip.

Fixed Body Skip IV

*Only moving part is door, that forms its own dumping chute.

*Suitable for fixed or rope guide.

*Hinged door at top can convert skip to a personnel & material cage.

*Skip requires less travel distance to open door.

*Skip place less stress on head frame, skip body, rope during dumping operation.

*Less spillage at door.

Fixed Body Skip V

Disadvantages

*Bail less construction requires adequate design & inspection during operation.

Fixed Body Skip I

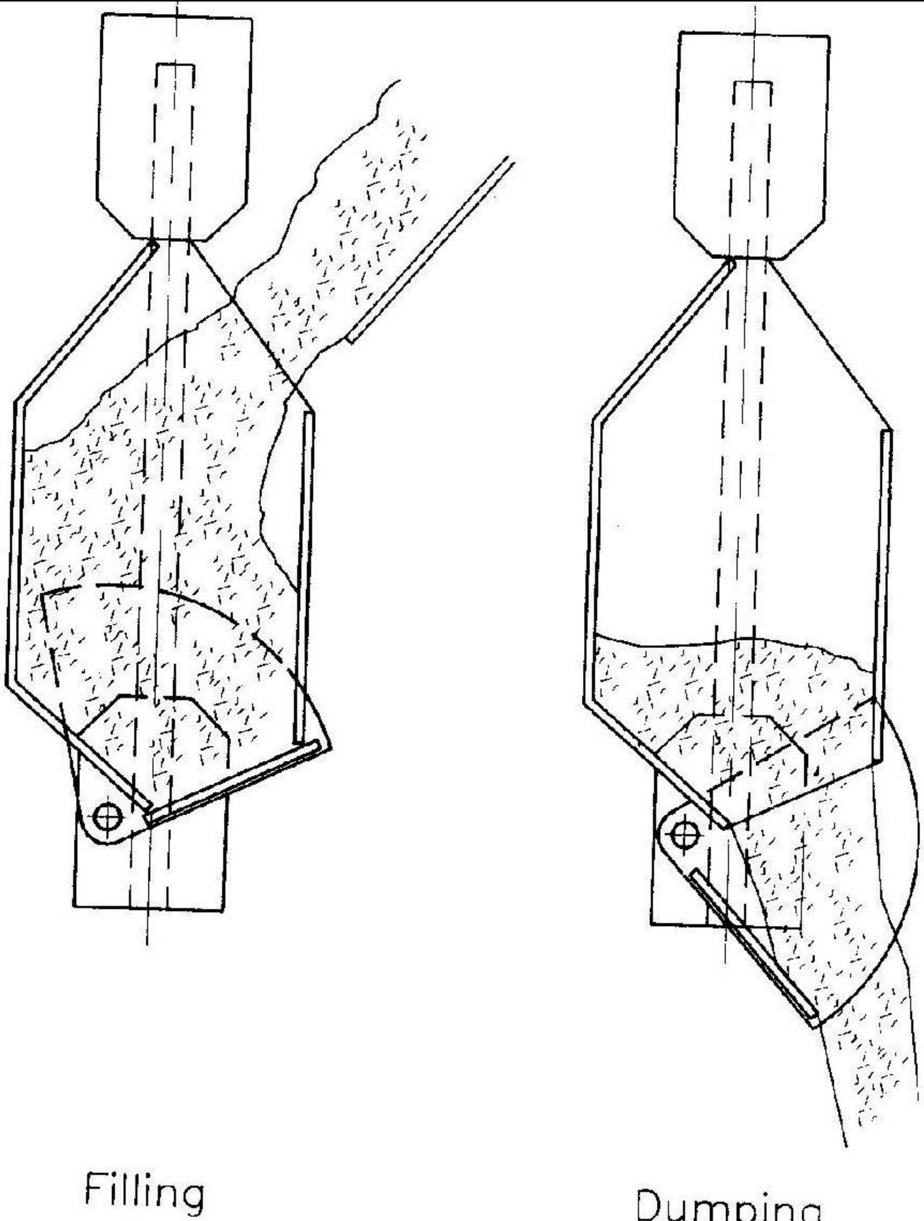

Fig. 17.5.9. Bottom dump skip, fixed body (Penning et al., 1977).

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Design Method Of Skip I

*Skip components are designed on basis of experience, practice & analytical method.

*Analytical method defines loading being places on various component of skip

*Type of loading considered are dynamitic loading, caused by filling, hoisting & dumping skip, static load factor of safety.

Conveyance & Accessories II

*Counter weight is used to balance with either skip or cage.

*Accessories consists of loading pockets, spill pockets & arrestor gear.

*Loading pocket improves efficiency of hoisting system by ensuring that optimum amount of material is fed to skip during each cycle.

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Fixed Body Skip V

Disadvantages

*Bail less construction requires adequate design & inspection during operation.

Skip Body I

*Skip body is designed as rectangular box with stiffeners.

*Type of materials, its thickness & stiffeners size & spacing are variables, they are to be designed based on duty of skip.

*Particular attention must be given to area on back side of skip, that receives impact & abrasion from loading.

*Skip bodies are lined with abrasion resistant (AR) material to increase life if skip.

Skip Body II

*Abrasion Resistant (AR) steel are member of high strength, low alloy family steels.

*Manganese steel are used because they have ability to work during service.

*"Elastomer" lining have advantage of reduced weight, less ore sticking & build up, corrosion resistant & ease of handling..

*Use of aluminum is also increasing, particularly in drum hoist application.

*Such skip has lower mass & allows higher pay load to be hoisted.

Skip Door I

*Both swing out body & fixed body skip have doors at bottom of skip.

*Doors must be designed to with stand impact of large piece falling on floor during loading.

*Skip door also has to stand static loading of a full skip as well as abrasion during unloading.

*With swing out body skip, door is hinged at back.

*There is possibility of spillage of ore in swing out door & body.

*Door spillage problem may be solved through use of rubber sheet, flexible material or modification of plate at door.

Skip Door I

*Fixed body skips have fewer problems with leakages or door closer.

*In case, skip is loaded on side opposite “dump Site” a vertical stop has to be placed for dump rollers to react against..

*Doors are made & lined with same material as used in body.

Cross Head I

*Cross head fasten bail to rope.

* Cross head must be designed to take all skip loads, including acceleration, impact & service factor.

* Cross head should not interfere with flow of muck from loading.

*Rope stretch in deep mines can be several feet & must be considered while designing cross head & bail frame.

Bail Frame I

*During hoisting, bail frame supports body pivot, prevents body from moving & support door.

*At time of dump, bail is subjected to horizontal loading thus subjected to both bending & axial loading.

Guide Roller I

*Purpose of guide roller is to guide conveyance for movement in shaft.

*Guide roller provide smooth ride, reduce dynamic loading between skip, guide & bunton during travel.

*Guide roller is selected based on adequate shaft clearance, cross section of skip.

*Guide roller tire needs frequent change.

*Tier size ranges from 152mm (6inch) to 356mm (14 inch)

*Tiers are rated on load per hour basis.

Guide Shoes I

*Guide Shoes are used in vertical conveyance to keep main guide frame from being worn away by guide.

*They are consumable item & are replaced when they are too worn for fresh use.

*Clearance between guide & guide shoes varies from 6 to 12mm.

*Guide shoes can be made from cast iron, mild steel, hardened steel, AR steel, brass, bronze.

*Amount of ware to be expected directly proportional to load on guide shoes & inversely proportional to hardness of material.

Guide Shoes II

*Guide shoe should be softer than guide material.

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Loading Station I

*At loading station, ore & waste are transferred from mine to skip.

*System & equipment can be very simple manually operated or complex & automated.

*Loading station can adopts measurement by volume or by weight.

*Criteria to be considered to design loading station are response time, accuracy of measurement device, operating & maintenance cost, amount of spillages,

*Loading station includes chute with control gate, measuring bin, skip feed chute with control gate.

Ore Bin (Pocket Chute) I

*Ore bin can be excavated in rock or be fabricated from steel & liner.

*Ore bin has a chute with control gate.

*Chute has a inclination of 45 degree to 55 degree.

*Chute gates are either pneumatically or hydraulically controlled.

*Chute gat can be of gullotine, ball & chain or redial type.

Measuring Bin (Pocket) I

*Measuring bin construction is similar to that of skip body.

*There are two type of fabricated measuring bin available, swing out body & fixed body.

*Fixed body is proffered because of its faster operation.

*In this there is no large mass to be moved.

*This can be easily actuated.

*Measuring bin (Loading) is furnished with pneumatic or hydraulic control gate.

Measuring Bin (Pocket) II

*Bottom of loading pocket is usually inclined at an angle of 60 degree or greater to ensure rapid loading & complete clean out.

*With balanced skip loading, two loading pockets are provided, one for each skip.

Skip Feed Chute I

*A feed chute can be used to direct material from measuring bin to skip.

*Angle of inclination of feed chute is usually 60 degree or greater.

*Clarence between chute & skip is maintained at about 75mm.

*Ratio of chute duscharge width to skip width should not exceed 0.7 for skip up to 1.3m width & 0.8 for skip width up to 1.8m

*To reduce spillage, side of chute should be slightly curved at tip.

Automatic Loading I

*Automatic loading of skip is used to improve efficiency of hoisting system & to reduce labour requirement.

*Efficiency improves by ensuring that each skip receives requirement load with out over loading or under loading.

*Man power is reduced when automatic loading is used in conjunction with automatic hoisting.

*For automatic loading system, following 3 control system must be included.

*Means of measuring weight or / & volume of material in measuring pocket.

Automatic Loading II

*Means of opening & closing feed chutes to fill empty measuring bin (pocket).

*A logic system to ensure that above operations are carried out in current sequence to prevent system failure.

*System failure could lead to double loading, dumping into an empty shaft.

*With automatic hoisting logic system is extended to operation of winder & other ancillary system.

Spill Bin (Pocket) I

*Despite of all precautions taken to prevent spillages during loading & dumping of skip, a certain amount of material falls below loading station.

*Spill can be confined to hoisting compartment & down to spill pocket.

*Normal amount of spills is about 1.5 to 5% of total hoisting.

*Large amount of spill is associated with manually operated system handling uncrushed material.

*Excess water in shaft or hoist material also increases spill.

Spill Bin (Pocket) II

*One method of handling spillage is to install deflector installed bellow skip.

*Deflectors in shaft is installed to direct spill to spill pocket, similar to loading pocket.

*Spill bins are to be cleaned on regular basic.

*In drum winder hoisting system, spill bin can be installed at any level, below loading bin.

*In case of friction winder, spill bin must be located below tail rope loop & second conveyance such as cage, used to handle spill.

Spill Bin (Pocket) III

*A ramp can be developed from lowest elevation to shaft bottom.

*In this system, spill falls directly at shaft bottom, from where, spill are mucked out by LHD.

*For friction winder system, it should be ensured that tail (balance) rope & cheese weight of guide rope should not be effected by build up of spill in shaft.

Cage I

*Cage are used primarily to handle personnel & material entering, exiting from mine.

*At some places, cage are also used to hoist loaded car with ore & waste.

*Basic design & construction of cage & skip are same.

*Cage is mainly an enclosed box, opened by door(s) & suspended from hoisting rope

*Dimension & capacity of cage are determined by quality, volume, weight & dimension of material & other item to be handled.

Counter Weight I

*In some balance hoisting system, a counter weight may be used with either cage or skip.

*Counter weight serves no purpose other than to provide balance weight for hoisting.

*Dimension & shape of counter weight is designed to fit in available space inside shaft.

Safety Device, Drum Winder I

*With drum hoist, both excessive over wind & rope breakages can occurs.

*In case of rope breakage, there should be provision to stop conveyance safety.

*A pair of safety dog is installed at top of conveyance at each guide.

*During travel tension in hoisting rope keeps dogs open.

*Should rope breaks or become slack, a large spring & lever to cause dig to penetrate guide & stop conveyance.

Safety Device, Friction Winder I

*In multiple rope friction hoist installation, chances of breaking ropes is very small.

*Provision of stoping over wind (over travel) has to be made in winding system.

*Arrester is used to stop over travel physically.

*Arrester consists of steel frame & wooden block supported in hoisting compartment & traps conveyance as it travel through arresting zone.

*These devi9ce is located at head frame as well as shaft bottom.

*They relay on friction between device & guide to decrease momentum.

Reference

*SME Mining Engineering Hand Book, 2nd Edition, Volume 2, Page 1658 to 1661,